

Smart Electricity Meter Anomaly Detection

Addressing Non-Technical Loss (NTL), Equipment Failures, and Field Dispatch Efficiency

Problem Overview:

Electric utilities face significant operational and financial losses due to Non-Technical Losses (revenue leakage, meter tampering, illegal connections) and unaddressed technical faults (calibration drift, frozen meters). Traditional detection relies on manual inspections or simple baseline thresholds that produce high false-positive rates and completely miss subtle edge cases like frozen meters or gradual calibration decay.

1. Key Industry Challenges

Revenue Loss & Fraud

Power theft through physical tampering and illegal line taps directly drains revenue from utility providers while introducing grid safety risks.

Undetected Hardware Faults

Meters exhibiting calibration drift or stuck zero-readings pass basic validation checks, leading to under-billing and inaccurate demand forecasting.

2. Deficiencies in Traditional Detection Methods

Generic Models Miss Critical Edge Cases: Standard unsupervised models (e.g., generic Isolation Forests) achieve near 0% detection on stuck/frozen meters, leaving massive blind spots in monitoring.

High False Positive Rates: Simple rule-based alerts fail to account for weather patterns, load-shedding, or consumer segment variations, resulting in wasted field inspection trips.

Lack of Root-Cause Granularity: Existing systems flag "anomalies" without distinguishing between criminal theft and technical degradation, complicating operational triage.

3. Target Objectives & Project Scope

The objective of this project is to develop an end-to-end, domain-specific Machine Learning architecture that processes raw interval telemetry, isolates distinct fault signatures, and prioritizes field audit dispatches.

Anomaly Vector	Impact on Utility	Target ML Capability
Meter Tampering	Direct revenue theft & security hazards	Reach ~98% detection recall via multi-feature scoring
Stuck Meter	Unbilled consumption (silent failure)	Eliminate blind spot (jump from 0% to ~80% recall)
Calibration Drift	Inaccurate customer billing & grid metrics	Double baseline detection rate via trend analysis